

|          |                |                  |         |
|----------|----------------|------------------|---------|
| NOP (00) |                | Cycles: 2        | Size: 1 |
| Implied  |                |                  |         |
| Cycle    | R/W            | Desc.            |         |
| -X.2     | Read <b>PC</b> | Store as OpCode. |         |
| 1.1      |                | Inc. <b>PC</b> . |         |
| 1.2      |                |                  |         |
| 2.1      |                |                  |         |
| 2.2      |                |                  |         |
|          | Read <b>PC</b> | Store as OpCode. |         |
| +X.1     |                |                  |         |

|            |                                      |                               |
|------------|--------------------------------------|-------------------------------|
| TCALL (01) | Cycles: 8                            | Size: 1                       |
| Implied    |                                      |                               |
| Cycle      | R/W                                  | Desc.                         |
| -X.2       | Read <b>PC</b>                       | Store as OpCode.              |
| 1.1        |                                      | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                       | Store as Operand.             |
| 2.1        |                                      |                               |
| 2.2        |                                      |                               |
| 3.1        |                                      |                               |
| 3.2        | Write to u8( <b>S</b> .L+0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                      |                               |
| 4.2        | Write to u8( <b>S</b> .L-1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                      | Dec. <b>SP</b> by 2.          |
| 5.2        |                                      |                               |
| 6.1        |                                      |                               |
| 6.2        | Read \$FFDE                          | Store as <b>PC</b> .L.        |
| 7.1        |                                      |                               |
| 7.2        | Read \$FFDF                          | Store as <b>PC</b> .H.        |
| 8.1        |                                      |                               |
| 8.2        | Read <b>PC</b>                       | Store as OpCode.              |
| +X.1       |                                      |                               |

|                                     |                  |                                                                             |
|-------------------------------------|------------------|-----------------------------------------------------------------------------|
| SET1 (02)                           | Cycles: 4        | Size: 2                                                                     |
| Direct Page Bit (Read/Modify/Write) |                  |                                                                             |
| Cycle                               | R/W              | Desc.                                                                       |
| -X.2                                | Read <b>PC</b>   | Store as OpCode.                                                            |
| 1.1                                 |                  | Inc. <b>PC</b> .                                                            |
| 1.2                                 | Read <b>PC</b>   | Store as Operand.                                                           |
| 2.1                                 |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>SPW</b> & <b>P</b> flag) << 3). |
| 2.2                                 | Read Address     | Store as Operand.                                                           |
| 3.1                                 |                  | Operand = (Operand & ~1)   1.                                               |
| 3.2                                 | Write to Address | Write Operand.                                                              |
| 4.1                                 |                  |                                                                             |
| 4.2                                 | Read <b>PC</b>   | Store as OpCode.                                                            |
| +X.1                                |                  |                                                                             |

|                          |                |                                                                             |
|--------------------------|----------------|-----------------------------------------------------------------------------|
| BBS (03)                 | Cycles: 5-7    | Size: 3                                                                     |
| Direct Page Bit Relative |                |                                                                             |
| Cycle                    | R/W            | Desc.                                                                       |
| -X.2                     | Read <b>PC</b> | Store as OpCode.                                                            |
| 1.1                      |                | Inc. <b>PC</b> .                                                            |
| 1.2                      | Read <b>PC</b> | Store as Operand.                                                           |
| 2.1                      |                | Inc. <b>PC</b> . Address = (Operand   (( <b>SPW</b> & <b>P</b> flag) << 3). |
| 2.2                      | Read Address   | Store as Operand.                                                           |
| 3.1                      |                | Set Jump to (Operand & (1 << 0)) == (1 << 0).                               |
| 3.2                      |                |                                                                             |
| 4.1                      |                |                                                                             |
| 4.2                      | Read <b>PC</b> | Store as Operand.                                                           |
| 5.1                      |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2).        |
| 5.2                      |                |                                                                             |
| 6.1                      |                | Address = <b>PC</b> + int8( Operand ).<br><b>PC</b> .L = Address.L.         |
| 6.2                      |                |                                                                             |
| 7.1                      |                | <b>PC</b> .H = Address.H.                                                   |
| 7.2                      | Read <b>PC</b> | Store as OpCode.                                                            |
| +X.1                     |                |                                                                             |

|                    |                |                                                                                       |
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| OR (04)            | Cycles: 3      | Size: 2                                                                               |
| Direct Page (Read) |                |                                                                                       |
| Cycle              | R/W            | Desc.                                                                                 |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                |                | Inc. <b>PC</b> .                                                                      |
| 1.2                | Read <b>PC</b> | Store as Operand.                                                                     |
| 2.1                |                | Inc. <b>PC</b> . Address = (Operand   (( <b>SPW</b> & P flag) << 3).                  |
| 2.2                | Read Address   | Store as Operand.                                                                     |
| 3.1                |                | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1               |                |                                                                                       |

|                 |                |                                                                                       |
|-----------------|----------------|---------------------------------------------------------------------------------------|
| OR (05)         | Cycles: 4      | Size: 3                                                                               |
| Absolute (Read) |                |                                                                                       |
| Cycle           | R/W            | Desc.                                                                                 |
| -X.2            | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1             |                | Inc. <b>PC</b> .                                                                      |
| 1.2             | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1             |                | Inc. <b>PC</b> .                                                                      |
| 2.2             | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1             |                | Inc. <b>PC</b> .                                                                      |
| 3.2             | Read Address   | Store as Operand.                                                                     |
| 4.1             |                | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 4.2             | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1            |                |                                                                                       |

|                   |                                                  |                                                                                       |
|-------------------|--------------------------------------------------|---------------------------------------------------------------------------------------|
| OR (06)           | Cycles: 3                                        | Size: 1                                                                               |
| Indirect X (Read) |                                                  |                                                                                       |
| Cycle             | R/W                                              | Desc.                                                                                 |
| -X.2              | Read <b>PC</b>                                   | Store as OpCode.                                                                      |
| 1.1               | Read ( <b>X</b>   (( <b>SPW</b> & P flag) << 3)) | Inc. <b>PC</b> .                                                                      |
| 1.2               |                                                  |                                                                                       |
| 2.1               |                                                  |                                                                                       |
| 2.2               |                                                  | Store as Operand.                                                                     |
| 3.1               |                                                  | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2               |                                                  | Store as OpCode.                                                                      |
| +X.1              |                                                  |                                                                                       |

|                           |                                                                       |                                                                                       |
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| OR (07)                   | Cycles: 6                                                             | Size: 2                                                                               |
| X-Indexed Indirect (Read) |                                                                       |                                                                                       |
| Cycle                     | R/W                                                                   | Desc.                                                                                 |
| -X.2                      | Read <b>PC</b>                                                        | Store as OpCode.                                                                      |
| 1.1                       |                                                                       | Inc. <b>PC</b> .                                                                      |
| 1.2                       | Read <b>PC</b>                                                        | Store as Operand.                                                                     |
| 2.1                       |                                                                       | Inc. <b>PC</b> .                                                                      |
| 2.2                       |                                                                       |                                                                                       |
| 3.1                       |                                                                       | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>SPW</b> & P flag) << 3).             |
| 3.2                       | Read Address                                                          | Store as Pointer.L.                                                                   |
| 4.1                       |                                                                       |                                                                                       |
| 4.2                       | Read ((Address + 1) & \$FF)  <br>(( <b>SPW</b> & <b>P</b> flag) << 3) | Store as Pointer.H.                                                                   |
| 5.1                       |                                                                       |                                                                                       |
| 5.2                       | Read Pointer                                                          | Store as Operand.                                                                     |
| 6.1                       |                                                                       | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 6.2                       | Read <b>PC</b>                                                        | Store as OpCode.                                                                      |
| +X.1                      |                                                                       |                                                                                       |



|           |                |                                                                                                        |
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| OR (08)   | Cycles: 2      | Size: 2                                                                                                |
| Immediate |                |                                                                                                        |
| Cycle     | R/W            | Desc.                                                                                                  |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                       |
| 1.1       |                | Inc. <b>PC</b> .                                                                                       |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                      |
| 2.1       |                | Inc. <b>PC</b> . <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                       |
| +X.1      |                |                                                                                                        |

|                                                |                  |                                                                                     |
|------------------------------------------------|------------------|-------------------------------------------------------------------------------------|
| OR (09)                                        | Cycles: 6        | Size: 3                                                                             |
| Direct Page to Direct Page (Read/Modify/Write) |                  |                                                                                     |
| Cycle                                          | R/W              | Desc.                                                                               |
| -X.2                                           | Read <b>PC</b>   | Store as OpCode.                                                                    |
| 1.1                                            |                  | Inc. <b>PC</b> .                                                                    |
| 1.2                                            | Read <b>PC</b>   | Store as Operand.                                                                   |
| 2.1                                            |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>SPW</b> & <b>P</b> flag) << 3).         |
| 2.2                                            | Read Address     | Store as Operand0.                                                                  |
| 3.1                                            |                  |                                                                                     |
| 3.2                                            | Read <b>PC</b>   | Store as Operand.                                                                   |
| 4.1                                            |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>SPW</b> & <b>P</b> flag) << 3).         |
| 4.2                                            | Read Address     | Store as Operand1.                                                                  |
| 5.1                                            |                  | Operand0  = Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 5.2                                            | Write to Address | Write Operand0.                                                                     |
| 6.1                                            |                  |                                                                                     |
| 6.2                                            | Read <b>PC</b>   | Store as OpCode.                                                                    |
| +X.1                                           |                  |                                                                                     |

|                      |                         |                                                                         |
|----------------------|-------------------------|-------------------------------------------------------------------------|
| OR1 (0A)             | Cycles: 5               | Size: 3                                                                 |
| Absolute Boolean Bit |                         |                                                                         |
| Cycle                | R/W                     | Desc.                                                                   |
| -X.2                 | Read <b>PC</b>          | Store as OpCode.                                                        |
| 1.1                  |                         | Inc. <b>PC</b> .                                                        |
| 1.2                  | Read <b>PC</b>          | Store as Address.L.                                                     |
| 2.1                  |                         | Inc. <b>PC</b> .                                                        |
| 2.2                  | Read <b>PC</b>          | Store as Address.H.                                                     |
| 3.1                  |                         | Inc. <b>PC</b> .                                                        |
| 3.2                  | Read (Address & \$1FFF) | Store as Operand.                                                       |
| 4.1                  |                         | Bit = Address >> 13.<br><b>C</b> flag  = ((Operand & (1 << Bit)) != 0). |
| 4.2                  |                         |                                                                         |
| 5.1                  |                         |                                                                         |
| 5.2                  | Read <b>PC</b>          | Store as OpCode.                                                        |
| +X.1                 |                         |                                                                         |

|                                 |                  |                                                                                                             |
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| ASL (0B)                        | Cycles: 4        | Size: 2                                                                                                     |
| Direct Page (Read/Modify/Write) |                  |                                                                                                             |
| Cycle                           | R/W              | Desc.                                                                                                       |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                                                                                            |
| 1.1                             |                  | Inc. <b>PC</b> .                                                                                            |
| 1.2                             | Read <b>PC</b>   | Store as Operand.                                                                                           |
| 2.1                             |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>SPW</b> & P flag) << 3).                                        |
| 2.2                             | Read Address     | Store as Operand.                                                                                           |
| 3.1                             |                  | <b>C</b> flag = (Operand & \$80). Operand <= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 3.2                             | Write to Address | Write Operand.                                                                                              |
| 4.1                             |                  |                                                                                                             |
| 4.2                             | Read <b>PC</b>   | Store as OpCode.                                                                                            |
| +X.1                            |                  |                                                                                                             |